

Table 78. LQFP48 - Mechanical data

Symbol	millimeters			inches ^(14.)		
	Min	Typ	Max	Min	Typ	Max
A	-	-	1.60	-	-	0.0630
A1 ^(12.)	0.05	-	0.15	0.0020	-	0.0059
A2	1.35	1.40	1.45	0.0531	0.0551	0.0571
b ^{(9.)(11.)}	0.17	0.22	0.27	0.0067	0.0087	0.0106
b1 ^(11.)	0.17	0.20	0.23	0.0067	0.0079	0.0090
c ^(11.)	0.09	-	0.20	0.0035	-	0.0079
c1 ^(11.)	0.09	-	0.16	0.0035	-	0.0063
D ^(4.)	9.00 BSC			0.3543 BSC		
D1 ^{(4.)(5.)}	7.00 BSC			0.2756 BSC		
E ^(4.)	9.00 BSC			0.3543 BSC		
E1 ^{(4.)(5.)}	7.00 BSC			0.2756 BSC		
e	0.50 BSC			0.1970 BSC		
L	0.45	0.60	0.75	0.0177	0.0236	0.0295
L1	1.00 REF			0.0394 REF		
N ^(13.)	48					
θ	0°	3.5°	7°	0°	3.5°	7°
θ1	0°	-	-	0°	-	-
θ2	10°	12°	14°	10°	12°	14°
θ3	10°	12°	14°	10°	12°	14°
R1	0.08	-	-	0.0031	-	-
R2	0.08	-	0.20	0.0031	-	0.0079
S	0.20	-	-	0.0079	-	-
aaa ^{(1.)(7.)}	0.20			0.0079		
bbb ^{(1.)(7.)}	0.20			0.0079		
ccc ^{(1.)(7.)}	0.08			0.0031		
ddd ^{(1.)(7.)}	0.08			0.0031		

Notes:

1. Dimensioning and tolerancing schemes conform to ASME Y14.5M-1994.
2. The Top package body size may be smaller than the bottom package size by as much as 0.15 mm.
3. Datums A-B and D to be determined at datum plane H.
4. To be determined at seating datum plane C.
5. Dimensions D1 and E1 do not include mold flash or protrusions. Allowable mold flash or protrusions is "0.25 mm" per side. D1 and E1 are Maximum plastic body size dimensions including mold mismatch.
6. Details of pin 1 identifier are optional but must be located within the zone indicated.
7. All Dimensions are in millimeters.
8. No intrusion allowed inwards the leads.
9. Dimension "b" does not include dambar protrusion. Allowable dambar protrusion shall not cause the lead width to exceed the maximum "b" dimension by more than 0.08 mm. Dambar cannot be located on the lower radius or the foot. Minimum space between protrusion and an adjacent lead is 0.07 mm for 0.4 mm and 0.5 mm pitch packages.